

Python Application Programming

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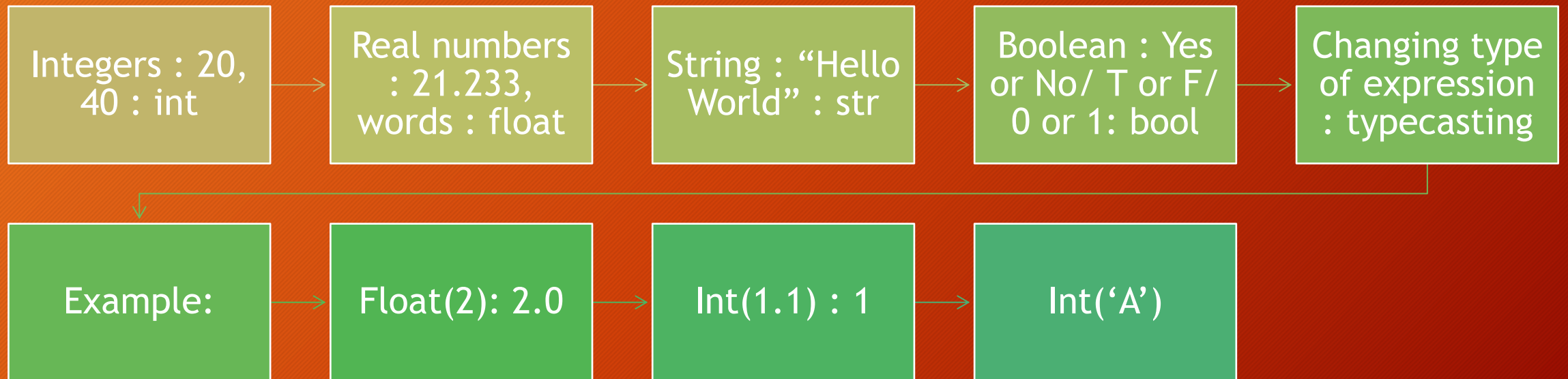
Introduction

- B.Tech Cloud Computing : 2014 - 2018
- Program Manager DRDO: 2017 - 2018
- MSCS Data Analytics IIT Chicago USA: 2019 - 2020
- Senior Technical Support Representative Codepath.org: 2020 - 2023
- Program Manager Codepath.org: 2021 - 2022
- Quantum Advisor PSA Indian: 2024 - Present
- Distinguish Scientist Niti Ayog: 2023
- 14 Scopus Papers , 4 Indian Patents filed, 2 UK Patents Granted
- Research Interests: Data Science, Data Analytics, Quantum Computing, Prompt Engineering

Requirements

- 2 Assignments
- 3 Quizzes
- Capstone Project
- Term 1
- Term 2
- End Term
- Discussion Forum

Types



Expression

- Expression: Type of operation the computer performs
- Example: $43+60+16+41 = 160$
- Numbers here are called operands
- Addition symbol are called operators
- For integer division we use $//$: $26//5 = 5$

Variables

```
My_variable = 1
```

```
X = 40 + 60
```

```
X:100
```

- `Type(x):float`

Strings

- Statements
- Integers
- Set of numbers
- Special characters

M	i	c	h	a	e	l		J	a	c	k	s	o	n
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

Name[0]:M Name[6]:l

Strings Continuation

- \ are meant to proceed escape sequences
- escape sequences are strings that are difficult to input

```
print(" Michael Jackson\n is the best" )
```

```
Michael Jackson  
is the best
```


List and Tuples

Tuples are basically rows in SQL.

Tuples are ordered sequence.

Example: `tuple1 = ('disco', 10, 1.0)`

- `tuple1[0] : 'disco'`

We can concatenate the tuples

Example: `tuple2 = tuple1+(50)`

- `tuple2: ('disco', 10, 1.0, 50)`

Slicing : `tuple2[0:3]`

Tuples are immutable : we can't change them

Nesting Tuple

